

Ticket System v3

Complete manual — project rundown & user guide

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1) Project rundown (technical)

1.1 Purpose

Ticket System v3 is a full-stack **service desk / ticketing** application for submitting requests, assigning work to teams and personnel, tracking **SLA** targets, managing transfer approvals for items that need higher-level handling, managing **KPI** maintenance tasks, and collecting **customer verification and feedback** after resolution.

1.2 Technology stack

Layer	Technology
Application	Next.js 16 (App Router), React 19 , Tailwind CSS 4
API	Next.js Route Handlers (src/app/api/**)
Database	PostgreSQL with Prisma ORM
Authentication	NextAuth.js — local credentials and optional Google OAuth
Email	SMTP (e.g. Brevo) for resolution and verification flows

Layer	Technology
Realtime / UX	Socket.IO dependencies, role-based shells, responsive sidebars, notification polling

1.3 Repository layout (high level)

- **src/app/** — Pages and API routes (App Router).
- **src/components/** — Shared UI (navigation, portal shell, dashboards).
- **src/lib/** — Business logic: auth, SLA, email, ticket actions, portal accounts.
- **prisma/** — Schema, migrations, seed.
- **ecosystem.config.cjs** — **PM2** process definition for production on Node hosts.
- **docs/** — User manuals, staff SOP, and Mermaid flowcharts.

1.4 Current implementation status

- Live UI branding is centralized as **AGCTek Help Desk**.
- Customer portal navigation now includes **Dashboard, Active Tickets, Knowledge Base**, and **Settings**; ticket creation remains available through dashboard actions or direct URL **/tickets/new**.
- Ticket intake supports screenshot upload/paste/drag-and-drop with stored request screenshots visible from ticket detail/workspace.
- Operational escalation is now a **Request for transfer** approval workflow. The stored ESCALATED status is displayed as **Transfer pending**.
- KPI/task recurrence supports **Daily, Weekly, Monthly**, and **Quarterly** cycles; quarterly periods are implemented as 4-month cycles.
- Account features include profile images, security changes, account action requests, and Admin/SuperAdmin review of pending requests.
- Admin reporting at **/reports** is implemented for transfer posture, queue mix, recent closures, and SLA summary.

1.5 Core concepts

- **Ticket** — Single customer issue with status (Open → In progress → For confirmation → Closed, plus Pending info and Transfer pending as needed).
- **Priority** — Includes a “**Set Priority Level**” (**UNSET**) default until staff sets **Low / Medium / High / Urgent**; work typically cannot move to **In progress** until priority is set (policy enforced in API/board).
- **SLA** — First-response and resolution due times derive from **SlaPolicy** per priority; Admin-triggered sweep scans unresolved tickets and refreshes reports. Transfer/escalation movement is handled through the approval workflow rather than automatic sweep escalation.
- **Teams & agents** — Tickets route to teams; **Personnel** work assigned queues from **/agent**.
- **Transfer pending** — User-facing label for tickets awaiting approval of a personnel-submitted transfer request.

2) Architecture & data

2.1 Request flow (simplified)

1. User authenticates → session carries **role** and **email**.
2. Customer submits ticket → **POST /api/tickets** creates row, SLA timestamps, activity log.
3. Staff manage lifecycle → **PATCH /api/tickets/[id]** (status, priority, more-information requests, transfer requests, resolution).
4. On **Resolved**, system may email the **requestor inbox** (SMTP); customer **verifies** and submits **rating** where configured.
5. If assigned personnel cannot continue a ticket, they submit **Request for transfer** from **/agent/tickets/[id]** to an Admin/SuperAdmin reviewer; approval moves it into the transfer-pending/assignment flow.

2.2 Notification email routing

- **Google sign-in:** notification and verification emails target the **Google account email** (authProvider: google).
- **Portal / credentials:** requestorEmail on the ticket aligns with the **portal account work email** stored in the database (with safeguards); synthetic portal identities may require an explicit notification email on intake.

2.3 Pending resolution confirmation (customers)

If a customer has a **resolved** ticket that still needs **verification and mandatory rating** (no feedback record yet), **new ticket creation may be blocked** until that flow is completed. The UI surfaces a **notification** and links to the **verification** path (e.g. /tickets/[id]/verification).

3) Authentication & roles

3.1 Sign-in

- **URL:** /signin
- **Methods:** **local credentials** (username or email + password) and **Google** (if configured).

3.2 Role resolution (typical)

1. Portal account role from the database when an account exists.
2. Env lists **ADMIN_EMAILS** / **AGENT_EMAILS** for bootstrap mapping.
3. Default **Customer** where no other rule applies.

3.3 Role summary

Role	Typical access
Customer	Dashboard (/), Active Tickets , Create ticket , knowledge base, verification/rating, /admin/account.
Personnel	/my-requests, Ticket and Task boards (/agent), insights, optional assignment board.

Role	Typical access
Admin / SuperAdmin	Ticket dashboard, create requests, personnel, all board tabs, priority alerts, task definitions, insights, SLA sweep.

Head / privileged portal flags may grant coordination powers per implementation.

SuperAdmin can create staff portal accounts, manage roles/designated companies, sync roster entries, and assign staff color tags. Admin users can manage within their designated company scope where enabled.

4) Deployment & operations

4.1 Environment

Copy `.env.example` to `.env`. Key groups: `DATABASE_URL`, **NextAuth** secret and URL, optional **Google** client settings, **SMTP (Brevo)** for mail, `APP_BASE_URL / NEXTAUTH_URL` for correct email links, optional admin email lists.

4.2 Database

- **Prisma:** `npx prisma migrate deploy` *or* `npx prisma db push` (depending on how the environment was baselined).
- **Helpdesk Google Form CSV (Insights task metrics):** after deploy, upload the sheet export once per environment — `npm run db:apply-helpdesk-csv path/to/"IT SALF - HELPDESK.csv"` (spreadsheet **Completed** is counted as **For confirmation** alongside **Closed**).
- **Seed (optional):** `npm run db:seed` for demo data.
- **KPI snapshots/backfills:** `npm run db:backfill-kpi-snapshots`, `npm run db:apply-kpi-sheet`, and related range/daily scripts are available for metrics data repair/import.
- One-off: `npx tsx scripts/ensure-unset-priority-data.ts` if using **UNSET** priority and SLA rows need ensuring after `db push`.

4.3 Production (Node + PM2)

- Build: `npm run build`
- Start: `pm2 start ecosystem.config.cjs` (or `npm run pm2:start`)
- Reload after deploy: `npm run deploy:pm2` or `pm2 restart ticket_system_v3 --update-env`
- See repository `DEPLOY_NGINX.md`, `DEPLOY_NGINX_WINDOWS.md`, or `DEPLOY_CPANEL.md` for host-specific steps.

5) Customer user guide

5.1 First steps

1. Open **/signin** (legacy **/customer/signin** redirects here; **/signup** when self-registration is enabled).
2. Use the account your organization provided, or register per policy.
3. After sign-in, the **Dashboard** (/) shows a kanban of your tickets.

Portal navigation

Area	Path
Dashboard	/
Active Tickets	/my-tickets
Knowledge Base	/tickets/knowledge
Settings	/tickets/knowledge#settings

Use dashboard actions or **/tickets/new** to create requests. Use the **notification bell** for updates and verification reminders.

5.2 Submit a ticket

1. Go to **/tickets/new**.
2. Complete **department / business unit, name**, and **issue** description.
3. Attach screenshots if helpful. The intake form supports upload, paste, and drag-and-drop; up to **15 images, 5MB each**.
4. If you use **Google**, emails for resolution/verification go to your **Google address**. If you use a **portal** account, notification email follows your **registered work email** (optional override only when it matches your portal email).
5. Submit; track from **/my-tickets** or the dashboard.

5.3 Track and participate

- Open a ticket from **Active Tickets**, the dashboard kanban, or a shared link.
- Read the **activity timeline** and **conversation** thread.
- Dashboard columns group work as **Open**, **In progress**, and **For confirmation** (resolved, awaiting your verify/rate).
- If **new requests are blocked** because a **resolved** ticket awaits your **confirmation and rating**, open the **notification** or the linked **verification** page and complete the steps.

5.4 Resolution, verification, and rating

1. When staff marks the ticket **Resolved**, you may receive email with verify / reject links.
2. In the portal, complete **verification** and **star rating / feedback** when prompted.
3. Ratings of **3 stars or below** require written feedback before closure.
4. Until mandatory steps are done, **creating another ticket** may be disabled **policy**.

5.5 Ticket statuses (customer view)

Status	Meaning (plain language)
Open	Received; queued / triaged.
In progress	Actively worked.
Pending info	Waiting on information (implementation may vary).
Transfer pending	A transfer request is waiting for Admin/SuperAdmin approval.
Resolved	Solution delivered; you should verify and rate.
Closed	Closed after confirmation / feedback per policy.

5.6 My account

Open `/admin/account`. The **Profile** tab supports profile image upload/framing (**PNG/JPG/WEBP**, up to **10MB**) and local bio notes. The **Security** tab changes username, email, and password; successful security changes sign you out and require a fresh sign-in. Submit suspension, deletion, or password reset requests for admin review when needed.

6) Staff & personnel guide

6.1 Daily routine

1. Sign in at `/signin`.
2. **Personnel** land on `/agent`; personal ticket list is `/my-requests`.
3. Use board tabs: **Ticket Board** (`/agent?board=ticket`) and **Task Board** (`/agent?board=kpi`).
4. Review `/insights` — ticket metrics, task metrics, and (coordinators only) **My Task Management**.

6.2 Working a ticket

1. Confirm assignment and priority (**set priority** if still **UNSET** — otherwise board/API may block **In progress**).
2. Update status honestly (drag-and-drop where enabled).
3. Use **request more information** per policy (may log activity **without** forcing “pending info” status depending on configuration).
4. Use **Request for transfer** from `/agent/tickets/[id]` when the assigned personnel needs Admin/SuperAdmin approval to transfer ownership or escalate handling.
5. Resolve with clear **resolution notes**; customer receives email to **verify**.
6. Staff ticket workspace: `/agent/tickets/[id]`.

6.3 KPI / task hygiene

- Complete checklist items where you are the KPI card assignee, or where an individual sub-KPI is assigned directly to you.

- Admins/coordinators can assign individual sub-KPIs to other personnel from inside each Task Board card.
- Non-IT Project sub-KPIs can store **before** and **after** screenshot sets for evidence; uploads must be **JPEG or PNG**, up to **5 screenshots per slot**, no larger than **10MB** each.
- Respect recurrence boundaries (daily / weekly / monthly cycles); period boundaries use browser/reporting timezone in API calls.
- **Recurring tasks:** flat or **segmented** sub-KPI checklists; **Daily, Weekly, Monthly**, and **Quarterly** schedules; checklists reset on period rollover.
- **IT Project Implementation:** non-recurring; phases and per-sub-task due dates (**MM/DD/YYYY** in UI); record **actual date** when completing sub-tasks. **Delayed** column applies only to this pillar (late sub-task or actual after due); fully complete but late work stays in **Delayed**, not **Done**.
- Other recurring KPIs: late completion may show **Done** with a delayed indicator.
Quarterly task metrics currently use 4-month cycles and daily task metrics exclude Sundays from working-day averages.

6.4 Metrics & Reports (/insights)

Tab	Audience	Content
Ticket metrics	Admin, Personnel	SLA, volume, CSAT, charts (date range)
Task metrics	Admin, Personnel	Helpdesk / checklist pillar metrics by daily, weekly, monthly, or quarterly cadence
Task Management	Personnel coordinators	KPI definitions (when canAccessAssignmentBoard)

Admins define tasks on **Task Board** (/agent?board=kpi), not the Insights Task Management tab.

7) Admin & SuperAdmin guide

7.1 Oversight

- / — Ticket dashboard (on-duty, activity, priority stack).
- /admin/ticket-requests — Create requests on behalf of users.
- /admin/personnel — Personnel registry and assignment readiness.
- /admin/manual-assignment — Assignment board (drag to personnel lanes).
- /agent?board=company — Company board.
- /agent?board=ticket — Ticket board.
- /agent?board=kpi — Task board + **Task Definition** console.
- /admin/escalation-triggers — **Priority alerts** (escalation configuration).
- /insights — Metrics & reports.
- /process — Process / lifecycle reference.
- /reports — Reporting views as implemented.

- **/api/brand/logo** — Serves the configured brand logo from `BRAND_LOGO_PATH` or bundled fallback.

7.2 SLA sweep

- **POST /api/sla/sweep** (secured; Admin context) — scans unresolved tickets and refreshes reporting paths. Current transfer/escalation movement is not automatic; personnel submit transfer requests for reviewer approval.

7.3 Accounts

- **/admin/account-management** — Redirects to **/admin/personnel**. Portal account administration is handled in Personnel, including account creation, role/designated-company updates, roster sync, duplicate-agent reconciliation, assignment colors, and request review.
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8) Account security (My Account)

All roles with portal access use **/admin/account** (Security tab).

The account area includes **Profile**, **Security**, and **Billing** tabs. Billing is present as a UI area only; ticket operations are managed through the ticket and reporting pages.

Profile image

Upload a profile image from **/admin/account** → **Profile**. Supported formats are **PNG**, **JPG**, and **WEBP**, up to **10MB**. After upload, drag/zoom the framing and save it.

Change username

1. Security tab → new username, current password → **Update username**.
2. You are signed out; sign in with the new username.
3. Rules: length and character constraints shown in UI.

Change email / password

Similar pattern: confirm current password; after success, **sign in again** with new credentials.

Account requests

Suspension, deletion, or password reset requests may be submitted for admin review. Approved password reset requests reset the portal password to the configured default reset value.

9) Troubleshooting

Symptom	What to try
Cannot sign in	Verify credentials, caps lock, Google sign-in status if used, and account active state.
"Forbidden" or missing menu	Your role may not include that feature — contact Admin.
Cannot create new ticket	Complete pending resolved verification/rating on existing ticket.
Email link expired / invalid	Request staff to re-send resolution flow or open ticket in portal.
Cannot move ticket to In progress	Set priority above Set Priority Level (UNSET) .
Ticket shows Transfer pending	A selected Admin/SuperAdmin reviewer must approve the transfer request.
Task stuck in Delayed (IT Project)	Complete overdue sub-tasks or correct actual/due dates.
Cannot edit KPI checklist	Confirm you are the assignee; IT Project requires per-sub-task actual dates.
Database / deploy errors	Check DATABASE_URL , run Prisma migrate or push, rebuild, restart PM2.

10) Quick reference — URLs

Path	Description
/signin	Sign in
/signup	Self-registration (if enabled)
/	Customer dashboard or admin ticket dashboard
/my-tickets	Customer active tickets
/my-requests	Personnel ticket dashboard
/tickets/new	New ticket
/tickets/knowledge	Knowledge base
/tickets/[id]	Ticket detail
/tickets/[id]/verification	Resolution verification
/tickets/[id]/rate	Rating / feedback
/agent	Staff boards (default ticket view)
/agent?board=ticket	Ticket board
/agent?board=kpi	Task board

Path	Description
/agent?board=company	Company board (admin)
/agent/tickets/[id]	Staff ticket workspace
/admin/manual-assignment	Assignment board
/admin/ticket-requests	Admin create requests
/insights	Metrics & reports
/process	Process / lifecycle info
/reports	Admin/SuperAdmin executive reporting
/admin/personnel	Personnel admin
/admin/escalation-triggers	Priority alerts
/admin/account-management	Redirects to Personnel portal-account admin
/admin/account	My Account & security
/api/brand/logo	Current brand logo asset

End of document.